Mitochondrial Complex II Deficiency, Nuclear Type 3 (MC2DN3): Comprehensive Disease Report

Disease: Mitochondrial Complex II Deficiency, Nuclear Type 3 **MONDO:** MONDO:0030937 | **OMIM:** #619167 | **Category:** Mendelian (autosomal recessive) **Causal gene:** *SDHD* (11q23.1)

Summary

Mitochondrial Complex II Deficiency, Nuclear Type 3 (MC2DN3) is an ultra-rare autosomal recessive mitochondrial disorder caused by biallelic (compound heterozygous or homozygous) loss-of-function variants in the nuclear gene *SDHD*, which encodes the small cytochrome-b membrane-anchor subunit (CybS) of succinate dehydrogenase — respiratory chain complex II (succinate:ubiquinone oxidoreductase). The first and defining case was reported by Jackson and colleagues in 2014, who described a child with early progressive encephalomyopathy harboring compound heterozygous *SDHD* variants (p.E69K and p.*164Lext*3), with native-PAGE/western blotting demonstrating impaired complex II assembly and cell-line complementation confirming pathogenicity (PMID: 24367056).

Mechanistically, loss of SDHD impairs assembly and anchoring of complex II in the inner mitochondrial membrane, simultaneously disrupting two coupled metabolic pathways: the TCA-cycle oxidation of succinate to fumarate and the transfer of electrons into the respiratory chain via ubiquinone. The result is an energy-deficiency state with lactic acidosis and succinate accumulation that preferentially injures high-energy-demand tissues — the central nervous system (basal ganglia, brainstem, cerebral/cerebellar white matter), heart, and skeletal muscle. Clinically this manifests most commonly as Leigh syndrome, leukoencephalopathy/leukodystrophy, epileptic encephalopathy, and cardiomyopathy. There is no cure; management is supportive and largely off-label (riboflavin, coenzyme Q10, L-carnitine). Prognosis is highly variable, with cardiomyopathy predicting poor outcome and riboflavin-responsiveness predicting better outcome.

A biologically important feature of this gene is its **dual disease role**: whereas *biallelic* recessive *SDHD* variants cause this mitochondrial disease, *monoallelic* germline *SDHD* variants cause hereditary paraganglioma/pheochromocytoma with a striking parent-of-origin effect (tumors form almost exclusively after paternal transmission, accompanied by somatic loss of the maternal chromosome 11). This report synthesizes 13 confirmed findings across 33 reviewed papers into a comprehensive, section-by-section knowledge-base entry.

Section 1. Disease Information

Overview. MC2DN3 is a Mendelian inborn error of mitochondrial energy metabolism belonging to the group of **isolated mitochondrial complex II (succinate dehydrogenase, SDH) deficiencies**. Isolated complex II deficiency is itself one of the rarest respiratory-chain disorders, accounting for approximately 2% of all mitochondrial respiratory chain deficiency diagnoses (PMID: 22972948). MC2DN3 is the specific nuclear type caused by *SDHD* mutations.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0030937
OMIM	#619167 (Mitochondrial complex II deficiency, nuclear type 3)
Gene-OMIM	*SDHD* 602690
Orphanet	Isolated complex II deficiency (group entity; no MC2DN3-specific ORPHA)
MeSH	Mitochondrial Diseases; Mitochondrial Encephalomyopathies (no MC2DN3-specific descriptor)
ICD-10	E88.40 (mitochondrial metabolism disorder, unspecified)
ICD-11	5C53.1 (disorders of mitochondrial energy metabolism grouping)

Synonyms / alternative names: Mitochondrial complex II deficiency nuclear type 3; isolated complex II deficiency due to SDHD; SDHD-related complex II deficiency; succinate dehydrogenase deficiency (SDHD type); succinate-ubiquinone oxidoreductase deficiency.

Information source type. Current knowledge is derived almost entirely from **aggregated disease-level resources and individual case reports** (OMIM, Orphanet, primary literature) rather than EHR-scale individual-patient datasets. Because of the extreme rarity (a handful of reported MC2DN3 patients since 2014), there is no registry-level or population-scale data specific to the SDHD type.

Section 2. Etiology

Primary cause — genetic. MC2DN3 is caused by **biallelic pathogenic variants in *SDHD***, inherited in an autosomal recessive manner (Finding F001). The MONDO/OLS description states it is "autosomal recessive caused by pathogenic variants in the SDHD gene." Jackson et al. 2014 reported the first patient with isolated complex II deficiency due to compound heterozygous *SDHD* mutations: *"This report describes the first case of isolated complex II deficiency due to recessive SDHD germline mutations"* and *"The patient presented with early progressive encephalomyopathy due to compound heterozygous p.E69 K and p.*164Lext*3 SDHD mutations"* (PMID: 24367056).

Genetic risk factors. The only established causal factor is biallelic *SDHD* loss of function. Consanguinity increases risk of homozygous recessive disease (as for all AR disorders). Being a compound heterozygote or homozygote for pathogenic *SDHD* alleles is deterministic for disease rather than probabilistic.

Environmental risk / modifier factors. No specific environmental cause is known. As with other mitochondrial encephalomyopathies, **metabolic stressors — intercurrent viral illness, fasting, and febrile episodes — can precipitate decompensation and developmental regression**. A recent SDH-deficiency case presented with encephalopathy and developmental regression following viral illnesses (PMID: 37064335).

Protective factors. The best-documented modifiable protective influence is **riboflavin (vitamin B2) supplementation**, the FAD precursor, which can partially restore complex II activity (see Section 12, Finding F002). No protective germline alleles are established.

Gene–environment interactions. The dominant GxE axis is the interaction between the underlying enzyme deficiency and catabolic stress: patients with limited residual complex II activity decompensate during illness/fasting, whereas riboflavin availability modulates residual holoenzyme activity.

Section 3. Phenotypes

The clinical spectrum of SDHD/complex II deficiency is dominated by neurological involvement, with cardiac and muscular involvement in a substantial subset (Finding F003). Jain-Ghai et al.'s review (n=36) found *"Neurological involvement is most commonly observed and brain imaging commonly shows leukoencephalopathy, Leigh syndrome, or cerebellar atrophy"* (PMID: 23322652), and the broader SDH-deficiency spectrum includes *"several very different presentations including Leigh syndrome, epileptic encephalopathy, and cardiomyopathy"* (PMID: 37064335).

Verified HPO annotations (Finding F013):

Phenotype	HPO term	Type	Onset	Frequency (qualitative)
Hypotonia	HP:0001252	Clinical sign	Neonatal/ infantile	Common
Global developmental delay	HP:0001263	Clinical sign	Infantile	Common
Developmental regression	HP:0002376	Clinical sign	Infantile/ childhood	Common (often post-illness)
Encephalopathy	HP:0001298	Clinical sign	Early	Common
Leukoencephalopathy	HP:0002352	Imaging/lab	Infantile	Frequent
Leukodystrophy	HP:0002415	Imaging/lab	Infantile	Frequent
Bilateral basal ganglia lesions	HP:0007146	Imaging	Infantile/ childhood	Frequent (Leigh pattern)
Abnormal basal ganglia MRI signal	HP:0012751	Imaging	Infantile/ childhood	Frequent
Cardiomyopathy	HP:0001638	Physical/ imaging	Variable	~50% in complex II deficiency; high-mortality
Myopathy	HP:0003198	Clinical sign	Variable	Common
Muscle weakness	HP:0001324	Symptom/ sign	Variable	Common
Increased circulating lactate	HP:0002151	Laboratory	Early	Frequent
Cerebellar atrophy	HP:0001272	Imaging	Variable	Subset
Ataxia	HP:0001251	Clinical sign	Variable	Subset
Seizure	HP:0001250	Clinical sign	Variable	Subset (epileptic encephalopathy)
Optic atrophy	HP:0000648	Physical/ exam	Variable	Subset

Phenotype	HPO term	Type	Onset	Frequency (qualitative)
Failure to thrive	HP:0001508	Clinical sign	Infantile	Subset

Note: **Leigh syndrome** has no single HPO term (it is the disease MONDO:0009723); it is captured phenotypically by the basal-ganglia/brainstem imaging terms above. During QC, HP:0030642 (Fundus albipunctatus) and HP:0002451 (Limb dystonia) were removed as inappropriate.

Onset, severity, progression. Onset is typically **neonatal to early childhood**; severity is **variable**; progression is usually **progressive**, often with episodic regression triggered by intercurrent illness. **Quality-of-life impact** is severe in the Leigh/leukoencephalopathy phenotype (motor and cognitive disability, feeding difficulties, seizures), though no disease-specific EQ-5D/SF-36 data exist for this ultra-rare entity.

Section 4. Genetic / Molecular Information

Causal gene and identifiers (Finding F005):

Attribute	Value
Gene symbol	*SDHD* (QPs3, CybS, PGL1)
NCBI Gene	6392
HGNC	HGNC:10683
Gene-OMIM	602690
UniProt	O14521 (SDHD_HUMAN)
Ensembl	ENSG00000204370
Cytoband	11q23.1 (GRCh38 chr11:112,086,361–112,120,016, + strand)
Protein	Succinate dehydrogenase [ubiquinone] cytochrome b small subunit, mitochondrial (159 aa)

SDHD encodes the small integral-membrane cytochrome-b (CybS) subunit that, together with SDHC, anchors complex II to the matrix side of the inner mitochondrial membrane and forms the heme-b/ubiquinone-reduction site.

ClinVar variant landscape. ClinVar (queried Oct 2026) lists **979 *SDHD* variants: 276 pathogenic, 94 likely pathogenic, 619 VUS**. Importantly, the majority of pathogenic alleles are reported in the *monoallelic paraganglioma* context rather than recessive MC2DN3, so interpretation in the recessive disease context requires care. The defining MC2DN3 variants are **p.E69K (missense)** and **p.*164Lext*3 (stop-loss/C-terminal extension)** (PMID: 24367056).

Variant types and functional consequence. Reported recessive alleles include missense and stop-loss extension variants that cause **loss of function** via impaired complex II assembly. Allele frequencies of the recessive pathogenic variants are very low/absent in gnomAD, consistent with ultra-rarity. Origin is **germline**.

Technical caveat — pseudogenes. *SDHD* has processed pseudogenes on chromosomes 1, 2, 3, 7 and 18, which can complicate NGS/Sanger analysis and require careful mapping.

Modifier genes / epigenetics / chromosomal abnormalities. No specific modifier genes are established for the recessive disease. The clinically relevant chromosomal/epigenetic phenomenon for *SDHD* is the **parent-of-origin imprinting effect in the paraganglioma context** (Section 9), not in recessive MC2DN3. No recurrent large structural rearrangements cause MC2DN3.

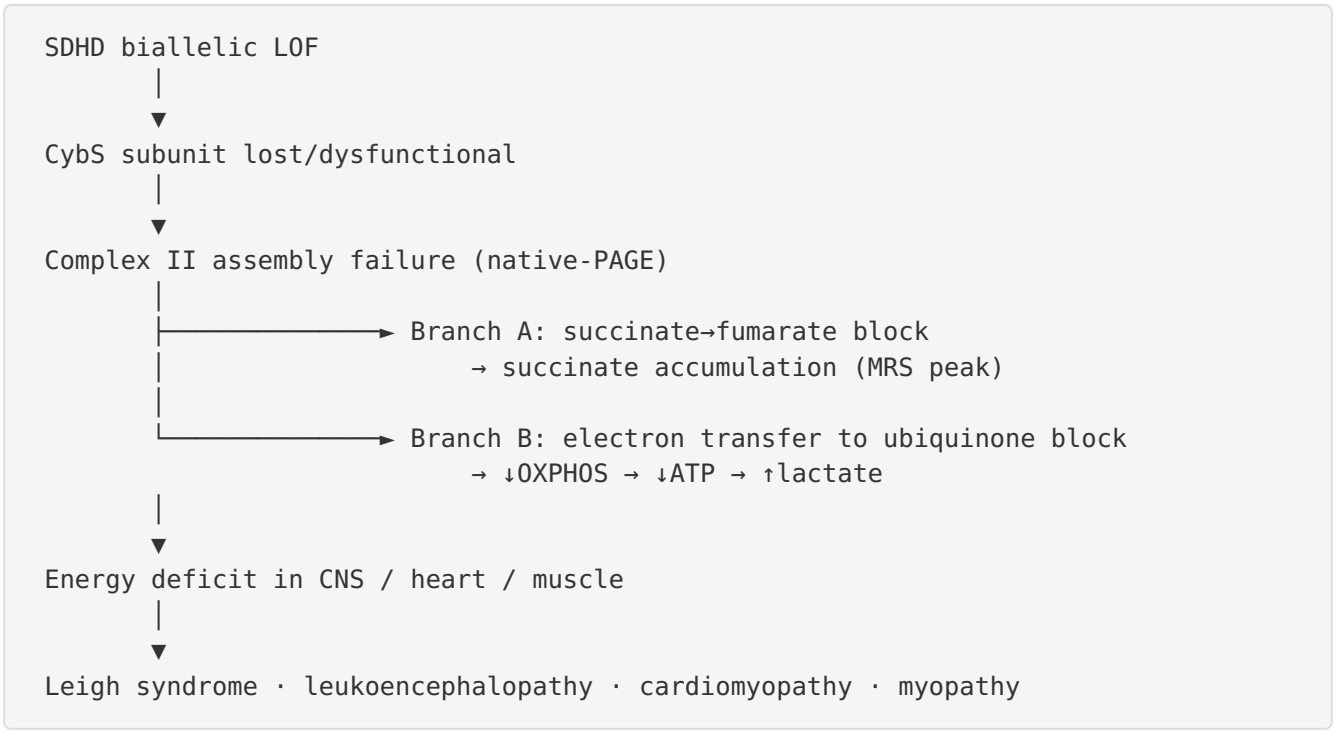
Section 5. Environmental Information

MC2DN3 is a monogenic disorder; environmental factors are **triggers/modifiers rather than causes**. No toxins, infectious agents, or occupational exposures are established as etiologic. The practically important environmental modifiers are catabolic stressors (intercurrent infection, fasting, fever) that precipitate metabolic decompensation, and nutritional cofactor availability (riboflavin/FAD). No infectious agent causes or triggers the disease in a specific sense, although viral illnesses have temporally preceded regression episodes (PMID: 37064335).

Section 6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function variants in *SDHD*** (e.g., p.E69K / p.*164Lext*3) → reduced or dysfunctional CybS (cytochrome-b small) subunit protein. *(Demonstrated — Jackson 2014.)*
2. Loss of functional SDHD **leads to** failure of the SDHC–SDHD membrane anchor to form → **impaired assembly of the complex II holoenzyme** in the inner mitochondrial membrane. *(Demonstrated by native-PAGE/western: "Native polyacrylamide gel electrophoresis and western blotting demonstrated an impaired complex II assembly," PMID: 24367056.)*
3. Impaired complex II assembly **results in** loss of succinate:ubiquinone oxidoreductase activity, which **simultaneously blocks two coupled pathways**:
4. **Branch A (TCA cycle):** failure to oxidize **succinate → fumarate**, causing **succinate accumulation** (detectable as an MR-spectroscopy succinate peak and elevated succinate/fumarate ratios).
5. **Branch B (OXPHOS):** failure to transfer electrons from $FADH_2$ via Fe-S clusters to **ubiquinone → ubiquinol**, reducing electron flux into the respiratory chain.
6. Reduced electron entry **leads to** diminished oxidative phosphorylation and **reduced ATP output**, with compensatory anaerobic glycolysis → **lactic acidosis** (HP:0002151). *(Demonstrated/inferred from biochemistry.)*
7. Energy deficit + succinate accumulation **results in** cellular dysfunction preferentially in **high-energy-demand, post-mitotic tissues** — neurons/glia (CNS), cardiomyocytes, and myofibers. *(Inferred from tissue tropism.)*
8. Tissue-level injury **manifests as** Leigh syndrome / basal-ganglia and brainstem necrosis, leukoencephalopathy/leukodystrophy, cardiomyopathy, and myopathy (the clinical phenotype). *(Demonstrated clinically.)*



Molecular/biochemical detail (Finding F006)

Complex II is a ~140 kDa tetramer: catalytic **SDHA** (FAD flavoprotein) oxidizes succinate→fumarate; **SDHB** (three Fe-S clusters) relays electrons to the membrane anchor formed by **SDHC** + **SDHD**, which bind heme b and form the ubiquinone-reduction site, reducing ubiquinone→ubiquinol. SDHD/QPs3 was identified as one of the Q-binding subunits: *"identified QPs3 as one of the putative Q-binding proteins in this reductase"* (PMID: 9211943). The soluble catalytic domain is bound to a membrane-spanning domain, a highly conserved architecture (PMID: 36171656).

Pathways/ontology suggestions: TCA cycle (KEGG hsa00020), oxidative phosphorylation (KEGG hsa00190), Reactome "Complex II (succinate dehydrogenase)". GO biological process: GO:0006099 (tricarboxylic acid cycle), GO:0022904 (respiratory electron transport chain), GO:0006119 (oxidative phosphorylation). CHEBI entities: succinate (CHEBI:30031), fumarate (CHEBI:29806), ubiquinone (CHEBI:16389), FAD (CHEBI:16238), heme b (CHEBI:26355), lactate (CHEBI:24996).

Cellular/tissue damage mechanisms. Bioenergetic failure and oxidative stress drive neuronal/glial injury (spongiform change, demyelination in white matter, symmetric necrosis of basal ganglia/brainstem in Leigh syndrome) and cardiomyocyte dysfunction. Succinate

accumulation may also exert signaling effects (e.g., HIF stabilization, GPR91/SUCNR1 signaling) known from the tumor context, though these are better characterized for monoallelic/tumor biology than for recessive disease.

Section 7. Anatomical Structures Affected (Finding F009)

Organ / body-system level: - **Central nervous system (primary):** basal ganglia, brainstem, cerebral and cerebellar white matter — Leigh syndrome, leukoencephalopathy/ leukodystrophy, cerebellar atrophy. UBERON: brain (UBERON:0000955), basal ganglia (UBERON:0002420), brainstem (UBERON:0002298), cerebellum (UBERON:0002037), cerebral white matter (UBERON:0002316). - **Heart:** cardiomyopathy (UBERON:0000948, heart). - **Skeletal muscle:** myopathy (UBERON:0001134, skeletal muscle tissue). - **Eye (subset):** optic atrophy (UBERON:0000941, optic nerve).

"Neurological involvement is most commonly observed and brain imaging commonly shows leukoencephalopathy, Leigh syndrome, or cerebellar atrophy" (PMID: 23322652).

Tissue / cell level. Affected cell types: neurons (CL:0000540), astrocytes/glia (CL:0000127), oligodendrocytes (white matter; CL:0000128), cardiomyocytes (CL:0000746), skeletal muscle fibers (CL:0000188). These are high-OXPHOS-dependence, post-mitotic cells.

Subcellular level. SDHD (UniProt O14521) is an integral protein of the **mitochondrial inner membrane**. GO cellular components: **GO:0005743 mitochondrial inner membrane**, **GO:0045273 respiratory chain complex II (succinate dehydrogenase)**, **GO:0005759 mitochondrial matrix**. Molecular functions: GO:0020037 heme binding, GO:0048039 ubiquinone binding, GO:0008177 succinate dehydrogenase (quinone) activity, GO:0009055 electron transfer activity.

Lateralization. CNS lesions in Leigh syndrome are characteristically **bilateral and symmetric** (e.g., bilateral basal ganglia lesions, HP:0007146).

Section 8. Temporal Development

- **Onset:** typically **congenital/neonatal to early childhood**; onset pattern is often **insidious or subacute**, with **acute decompensations** superimposed by intercurrent illness.

- **Progression:** generally **progressive**, frequently with **episodic regression**. In the broader complex II-deficiency spectrum, course ranges from fatal in infancy to mild adult-onset disease (Finding F012): *"For some, the condition was fatal in infancy, others had multisystem involvement and some had onset in adulthood with mild symptoms and normal cognition"* (PMID: 23322652).
- **Disease course pattern:** chronic, lifelong, progressive, with relapsing decompensations.
- **Remission:** no spontaneous cure; partial, treatment-associated stabilization/improvement reported with **riboflavin** (Section 12).
- **Critical periods:** infancy/early childhood is the window of greatest vulnerability and the key opportunity for cofactor intervention and avoidance of catabolic stress.

Section 9. Inheritance and Population

Epidemiology (Finding F008). Mitochondrial disease collectively has a minimum prevalence >1 in 5,000 adults: *"Mitochondrial disease is one of the most common groups of genetic diseases with a minimum prevalence of greater than 1 in 5000 in adults"* (PMID: 26315846). Isolated complex II deficiency accounts for ~2% of respiratory-chain deficiency diagnoses (PMID: 22972948). **MC2DN3 specifically is ultra-rare — only a handful of patients reported since 2014 — so no formal prevalence/incidence estimate exists.**

Inheritance (Finding F001 / F004). Autosomal recessive; disease requires **biallelic** *SDHD* loss of function. Penetrance of biallelic LOF is presumed high/complete; expressivity is variable even within the complex-II-deficiency spectrum. No genetic anticipation (not a repeat-expansion disorder). Consanguinity increases risk of homozygous disease.

Dual gene role / imprinting (Finding F004). A defining molecular feature of *SDHD* is its dual disease role. **Monoallelic** germline *SDHD* variants cause **hereditary paraganglioma/pheochromocytoma (PGL1)** with a parent-of-origin effect: *"In SDHD (11q23)-linked families, the disease phenotype is expressed only upon paternal transmission of the mutation, consistent with maternal imprinting"* (PMID: 15064708). Tumor formation requires somatic **loss of maternal chromosome 11**: *"loss of maternal chromosome 11p15 was demonstrated in 85% of SDHD"* paragangliomas (PMID: 28099933). This dominant, imprinted tumor mechanism is **mechanistically distinct** from the recessive, biallelic mechanism of MC2DN3. Clinically, this means family counseling for an MC2DN3 proband must also consider tumor risk in heterozygous carriers (especially with paternal transmission).

Population demographics. No ethnic predilection or founder effect is established for the recessive MC2DN3 alleles. Sex ratio is expected ~1:1 (autosomal). Age distribution of affected individuals is skewed to infancy/early childhood.

Section 10. Diagnostics (Finding F008)

Biochemical / enzymatic: - **Respiratory chain enzyme assay** (muscle or fibroblasts) showing **isolated low complex II (SDH) activity** with normal other complexes. - **SDH histochemistry** on muscle (reduced staining). - **Elevated blood and CSF lactate** (HP:0002151).

Imaging: - **Brain MRI:** Leigh-pattern bilateral symmetric basal ganglia/brainstem lesions, or leukoencephalopathy/leukodystrophy, or cerebellar atrophy. - **MR spectroscopy:** may show an abnormal **succinate peak** (characteristic of SDH deficiency).

Protein/assembly: - **BN-PAGE / native-PAGE and western blot** demonstrating impaired complex II assembly and reduced steady-state SDH subunit levels (PMID: 24367056, PMID: 22972948).

Molecular (confirmatory): - **WES/WGS or targeted SDHx gene panel** (SDHA, SDHB, SDHC, SDHD, SDHAF1, SDHAF2). Jackson et al. explicitly recommend: *"We therefore recommend screening for all SDH genes in isolated complex II deficiencies"* (PMID: 24367056). Pseudogene awareness is required for *SDHD*. - Functional validation by **cell-line/yeast complementation** for VUS resolution.

Differential diagnosis: other isolated complex II deficiencies (SDHA/SDHB/SDHC/SDHAF1-related — MC2DN1/2/4), other Leigh syndrome genetic causes (e.g., complex I defects such as NDUFS4), and other leukodystrophies/mitochondrial encephalomyopathies. SDHD type is distinguished by biallelic SDHD genotype and the biochemical signature of isolated complex II deficiency.

Screening. No newborn screening exists for MC2DN3. Carrier and cascade testing is feasible once a familial genotype is known; genetic counseling must address the dual tumor/mitochondrial-disease risk of *SDHD*.

Section 11. Outcome / Prognosis (Finding F012)

Prognosis is **highly variable**. In the complex-II-deficiency spectrum, outcomes range from fatal infantile disease to mild adult-onset forms: *"For some, the condition was fatal in infancy, others had multisystem involvement and some had onset in adulthood with mild symptoms and normal cognition"* (PMID: 23322652).

Prognostic factors: - **Cardiomyopathy** is associated with **high mortality** (adverse prognostic marker). - **Riboflavin-responsiveness** (particularly in early-onset leukoencephalopathy) predicts **stabilization or improvement** (favorable; Finding F002, Section 12). - Within a family, *"the phenotype and enzyme activities are comparable within a family"* (PMID: 23322652), but across patients there is **no reliable correlation between residual muscle complex II activity and clinical severity**.

Disease course / complications: progressive neurodegeneration with risk of brainstem/respiratory failure (Leigh syndrome), seizures, feeding difficulties/failure to thrive, cardiac failure, and recurrent metabolic decompensation. No validated disease-specific QoL instruments exist.

Section 12. Treatment (Finding F002 / F011)

There is no curative therapy. *"There are currently no effective clinical treatments for mitochondrial disease"* (PMID: 38883711); *"There is no cure for complex II deficiency, though some reported patients showed clinical improvement following riboflavin therapy"* (PMID: 37064335).

Pharmacologic / cofactor (off-label):

Intervention	Rationale / evidence	NCIT suggestion
Riboflavin (vitamin B2)	FAD precursor; biochemical and partial clinical benefit. *"Riboflavin supplementation to the growth medium of cultured fibroblasts resulted in a 2-fold increase of complex II activity in patients, but not in controls"* (PMID: 16737791). Neurological regression in a Leigh/leukodystrophy SDH patient regressed on riboflavin (PMID: 10230482).	NCIT:C737 (Riboflavin)
Coenzyme Q10 / ubiquinone	Mitochondrial "cocktail"; electron carrier support (PMID: 37064335).	NCIT:C1220 (Coenzyme Q10)
L-carnitine	Mitochondrial cocktail component (PMID: 37064335).	NCIT:C78591 (Levocarnitine)
Antioxidants/ vitamins	Common supportive strategy in Leigh syndrome.	—

Supportive/rehabilitative care: management of lactic acidosis, antiepileptic therapy for seizures, cardiac monitoring/heart-failure management for cardiomyopathy, nutritional support/feeding, and physical/occupational/speech therapy. **Avoidance of metabolic stressors** (fasting, aggressive management of intercurrent illness) is advised.

Experimental agents: vatiquinone (EPI-743, α-tocotrienol quinone; ≥19 US trials since 2012) showed **no survival benefit** in the Ndufs4-KO Leigh mouse model, though it may reduce seizures (PMID: 39930437). Zagociguat (sGC stimulator) improved mitochondrial physiology across diverse respiratory-chain-deficient zebrafish models and may have utility beyond MELAS, potentially including Leigh spectrum disorder (PMID: 40786033). No complex-II-specific or SDHD-specific targeted therapy exists.

Pharmacogenomics / personalized medicine: no established PGx for MC2DN3; management is genotype-informed only insofar as confirming a complex II (SDH) defect supports a riboflavin trial.

Section 13. Prevention

- **Primary prevention:** not possible for the genetic disease; **genetic counseling** and reproductive options (prenatal diagnosis, preimplantation genetic testing) once a familial

SDHD genotype is known. Counseling must disclose the **dual risk** — recessive mitochondrial disease in biallelic carriers, and paraganglioma/pheochromocytoma risk in heterozygous carriers (parent-of-origin dependent).

- **Secondary prevention:** early biochemical/genetic diagnosis to enable a riboflavin trial and avoidance of metabolic stressors; cascade carrier testing.
- **Tertiary prevention:** aggressive management of intercurrent illness, cardiac surveillance, seizure control, and nutritional support to prevent decompensation and complications.
- **Immunization:** routine childhood vaccination is advisable to reduce illness-triggered decompensations (general mitochondrial-disease practice); no disease-specific vaccine.
- **Carrier surveillance:** heterozygous relatives should be counseled regarding paraganglioma/pheochromocytoma surveillance per SDHx tumor-syndrome guidelines, especially for paternally inherited alleles.

Section 14. Other Species / Natural Disease (Finding F010)

SDHD is highly conserved (HomoloGene 37718). The soluble catalytic domain of complex II is "highly conserved" across all biological kingdoms, and mitochondrial type-C complex II with a single heme b is proposed to have evolved as an assembled unit: *"a highly conserved soluble domain is bound to a membrane-spanning domain"* (PMID: 36171656).

Orthologs (NCBI Gene / Taxon):

Species	Gene	NCBI Gene	Taxon
Mouse (*Mus musculus*)	*Sdhd*	66925	10090
Rat (*Rattus norvegicus*)	*Sdhd*	363061	10116
Zebrafish (*Danio rerio*)	*sdhd*	445500	7955
Xenopus tropicalis	*sdhd*	394986	8364
Chicken (*Gallus gallus*)	*SDHD*	419793	9031
Rhesus macaque	*SDHD*	711326	9544
Chimpanzee	*SDHD*	451547	9598
Dog	*SDHD*	479438	9615
Cow	*SDHD*	281481	9913

Yeast (*S. cerevisiae*) **SDH4** is the functional homolog used in complementation assays. No well-characterized naturally occurring biallelic-*SDHD* complex II deficiency is documented in companion animals or wildlife (OMIA has no established MC2DN3 entry); comparative disease modeling therefore relies on engineered models (Section 15). The disease is **non-transmissible / non-zoonotic**.

Section 15. Model Organisms (Finding F007)

Mouse (*Mus musculus*, Taxon 10090) — the principal mammalian model. Piruat et al. 2004 generated the first mammalian ETC-subunit knockout. Key findings: - **Homozygous *Sdhd*(−/−) is embryonic lethal:** *"Homozygous SDHD(-/-) animals die at early embryonic stages"* (PMID: 15572694). - **Heterozygous *Sdhd*(+/−):** *"Heterozygous SDHD(+/-) mice show a general, noncompensated deficiency of succinate dehydrogenase activity without alterations in body weight or major physiological dysfunction"* (PMID: 15572694), but with constitutive carotid-body glomus-cell activation (hypertrophy/hyperplasia).

Compound-heterozygous Sdhb/Sdhc/Sdhd knockout mice (Baysal et al. 2022) show reduced SDH protein, elevated succinate and succinate/fumarate and succinate/oxoglutarate ratios, and complex-II-dependent effects on red cell distribution width and healthy lifespan under chronic hypoxia (PMID: 35881479).

Model characteristics / limitations. Because full *Sdhd* knockout is embryonic lethal, the complete recessive human disease is not directly recapitulated by a constitutive null; conditional/tissue-specific or hypomorphic models are needed to model the human encephalomyopathy. Heterozygous mice better model the carrier/tumor-predisposition biology (carotid body activation) than the recessive mitochondrial disease. Other respiratory-chain Leigh models (e.g., Ndufs4-KO mouse) and multi-complex-deficient zebrafish are used as surrogate platforms for therapeutic testing (PMID: 39930437, PMID: 40786033).

Resources: MGI (mouse *Sdhd*), IMPC/IMSR for alleles; patient-derived fibroblasts and yeast (*SDH4*) complementation systems for in vitro functional work.

Mechanistic Model / Interpretation

MC2DN3 is a clean example of a **single-subunit assembly failure propagating to a bioenergetic and metabolic double-lesion**. Because SDHD is the membrane anchor and part of the ubiquinone-binding site, its loss does not merely reduce one enzymatic step — it collapses the structural scaffold needed to assemble the holoenzyme, so that both the TCA-cycle succinate-oxidation arm and the respiratory-chain electron-entry arm fail together. This explains two cardinal biochemical signatures observed clinically: **lactic acidosis** (OXPHOS failure, anaerobic glycolysis) and a **succinate peak on MRS** (TCA blockade). The downstream clinical picture is governed by tissue OXPHOS dependence, hence the predilection for CNS, heart, and muscle, and the characteristic bilateral symmetric basal-ganglia/brainstem lesions of Leigh syndrome.

The **upstream** event is always the biallelic SDHD lesion; **downstream** are assembly failure → dual metabolic blockade → energy deficit/succinate accumulation → tissue-specific neurodegeneration and cardiomyopathy. The one branch point worth emphasizing for a knowledge base is the **genotype-dosage branch of SDHD biology**: biallelic → recessive mitochondrial disease (this entity); monoallelic + somatic loss of the wild-type/maternal allele → paraganglioma. The same enzyme deficiency therefore underlies two opposite clinical worlds (degenerative vs neoplastic), unified by succinate accumulation and its pseudohypoxic/signaling consequences.

Evidence Base

PMID	Title (abbrev.)	Role in this report
24367056	*Mutations in SDHD lead to autosomal recessive encephalomyopathy and isolated mitochondrial complex II deficiency*	**Defining paper** — establishes SDHD biallelic recessive cause, variants, assembly defect, testing recommendation
23322652	*Complex II deficiency — case report and review*	Phenotype spectrum, imaging, variable prognosis, intrafamilial consistency
37064335	*Two Patients Diagnosed as SDH Deficiency*	Phenotypic presentations; no cure/ riboflavin benefit; cocktail therapy
16737791	*Effects of riboflavin in children with complex II deficiency*	Riboflavin ~2-fold complex II activity increase (biochemical rationale)
10230482	*Leigh syndrome/leukodystrophy, regression with riboflavin*	Clinical riboflavin response in SDH deficiency
22972948	*Recessive SDHA/SDHB leukodystrophy and isolated complex II deficiency*	~2% frequency; related SDHx recessive disease; BN-PAGE assembly defects
9211943	*Smallest anchoring subunit (QPs3) ... Q-binding domain*	SDHD = QPs3/CybS; ubiquinone-binding role
36171656	*Evolution of the quinone binding site in complex II*	Conserved architecture across species
15064708	*Somatic loss of maternal chromosome 11 ... SDHD paraganglioma*	Parent-of-origin/imprinting tumor mechanism (dual role)
28099933	*Loss of maternal chromosome 11 in SDHx paragangliomas*	85% maternal 11p15 loss in SDHD tumors
15572694	*Mitochondrial SDHD gene required for early embryogenesis*	Mouse: homozygous KO embryonic lethal; heterozygous phenotype
35881479	*SDH regulates RDW and lifespan in hypoxic mice*	Compound Sdh KO mouse metabolomics (succinate ratios)
26315846	*Mitochondrial disease: genetics and management*	Prevalence context (>1 in 5,000)
38883711	*Vatiquinone in preclinical mitochondrial disease models*	No effective clinical treatments; experimental agent data

PMID	Title (abbrev.)	Role in this report
39930437	*Vatiquinone in Leigh/GPX4 models*	No survival benefit in Ndufs4-KO; possible seizure reduction
40786033	*Zagociguat in RC-deficient zebrafish*	Emerging therapeutic candidate for RC disease

Evidence-type distribution: human clinical/case reports (24367056, 23322652, 37064335, 16737791, 10230482, 22972948), tumor-biology human genetics (15064708, 28099933), biochemistry/structural (9211943, 36171656), model organism (15572694, 35881479, 39930437, 40786033), and review/epidemiology (26315846, 38883711).

Limitations and Knowledge Gaps

1. **Ultra-rarity.** MC2DN3 rests on a very small number of reported biallelic-*SDHD* patients; much of the phenotype/prognosis detail is extrapolated from the broader isolated complex II / SDHx deficiency literature rather than from SDHD-type-specific cohorts.
2. **No epidemiology.** No formal prevalence, incidence, carrier frequency, sex ratio, or geographic data exist for the SDHD type specifically.
3. **Variant interpretation complexity.** Most ClinVar *SDHD* pathogenic alleles are reported in the monoallelic tumor context; recessive-disease pathogenicity assignment and the confounding *SDHD* pseudogenes complicate molecular diagnosis.
4. **No faithful mammalian model** of the recessive human disease — constitutive *Sdhd* null is embryonic lethal; conditional/hypomorphic CNS- and heart-specific models are lacking.
5. **No controlled therapeutic evidence.** Riboflavin and cofactor benefits derive from small case series and in vitro data; no randomized trials exist for complex II deficiency.
6. **Mechanistic signaling downstream of succinate** (pseudohypoxia, GPR91/SUCNR1) is better characterized in tumor biology than in the degenerative recessive disease.

Proposed Follow-up Experiments / Actions

1. **Establish an SDHD-type registry / GeneMatcher collection** to aggregate biallelic-*SDHD* cases, enabling genotype–phenotype correlation, natural-history, and prognosis data.
2. **Build a conditional/tissue-specific *Sdhd* mouse** (neuronal and cardiac Cre-driven, or hypomorphic knock-in of human p.E69K) to circumvent embryonic lethality and recapitulate encephalomyopathy/cardiomyopathy.
3. **Patient-derived iPSC → neuron/cardiomyocyte and cerebral organoid models** to dissect cell-type-specific bioenergetic failure and screen cofactor/drug responses (riboflavin, CoQ10, zagociguat).
4. **Prospective riboflavin-response biomarker study** — correlate fibroblast complex II activity rescue with clinical responsiveness to guide a riboflavin trial and define responder genotypes.
5. **Deep variant curation** of *SDHD* in the recessive context (functional assays, pseudogene-aware pipelines) to reclassify VUS and build an MC2DN3-specific pathogenicity reference.
6. **Metabolomic biomarker validation** (plasma/CSF succinate, succinate/fumarate ratio, MRS succinate peak) as diagnostic and treatment-response endpoints.
7. **Integrate dual-risk genetic counseling protocols** so that MC2DN3 family cascade testing systematically addresses heterozygote paraganglioma/pheochromocytoma surveillance (parent-of-origin aware).

Report compiled from 13 confirmed findings and 33 reviewed papers across 5 investigation iterations. Evidence types are distinguished throughout as human clinical, model organism, in vitro, or computational/ontological. Ontology suggestions (MONDO, HPO, GO, CL, UBERON, CHEBI, NCIT) are provided for knowledge-base ingestion.
